SEMIRING TENSOR NETWORKS

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

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1 Semiring

Definition 1. *A semiring is a tuple* $(S, \oplus, \otimes, 0, 1)$*, where* S *is a set,* $0, 1 \in S$*, and*

$$\oplus : S \times S \to S \quad , \quad \otimes : S \times S \to S \, ,$$

such that the following properties hold:

(i) $(S, \oplus, 0)$ *is a commutative monoid.*

(ii) $(S, \otimes, 1)$ *is a commutative monoid.*

(iii) $\otimes$ *distributes over* $\oplus$*, that is for any* $s_0, s_1, s_2 \in S$

$$s_0 \otimes (s_1 \oplus s_2) = (s_0 \otimes s_1) \oplus (s_0 \otimes s_2) \, .$$

(iv) 0 *annihilates* $\otimes$*, that is (check!)*

$$0 \otimes (s_0 \otimes s_1) = 0 \, .$$

We understand S as the set of coordinates in a tensor.

Definition 2. *A tensor with coordinates in* S *of order* $d \in \mathbb{N}$ *and leg dimensions* $m_{[d]} = (m_0, \dots, m_{d-1})$*, where* $m_k \in \mathbb{N}$ *for* $k \in [d]$*, is a map*

$$\tau : \bigtimes_{k \in [d]} [m_k] \to S \, .$$

A tensor network with coordinates in S *is a hypergraph* $\mathcal{G} = (\mathcal{V}, \mathcal{E})$*, where each node is decorated by a dimension* m_v *and each edge by a tensor of .*

2 Contractions

Definition 3. *Given a semiring* $(S, \oplus, \otimes, 0, 1)$ *and a tensor network* $\{\tau^e [X_e] : e \in \mathcal{E}\}$ *with coordinates in* S*, the contraction of the tensor network keeping the variables to a subset* $\mathcal{U} \subset \mathcal{V}$ *open is the tensor*

$$\langle \{\tau^e [X_e] : e \in \mathcal{E}\} \rangle^{(S, \oplus, \otimes, 0, 1)}$$

The distributive property (iii) of the semiring ensures, that we can compute contractions by sub-contractions.

3 Examples

3.0.1 Standard TN

Numerical TN use the semiring

$$(\mathbb{R}, +, \cdot, 0, 1) \, .$$

3.1 CSP

We take $S = \{0,1\}$ and interpret 0 as False, 1 as True. The

$$\vee\left[X_0, X_1\right] = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \left[X_0, X_1\right]$$

$$\wedge\left[X_0, X_1\right] = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \left[X_0, X_1\right]$$

The satisfiability of a constraint network $\{\tau^e\left[X_e\right] \,:\, e \in \mathcal{E}\}$ is the contraction

$$\bigvee_{x_{\mathcal{V}}} \bigwedge_{e\in\mathcal{E}} \tau^e\left[X_e = x_e\right] = \left\langle \{\tau^e\left[X_e\right] \,:\, e \in \mathcal{E}\} \right\rangle_{\varnothing}^{(\{0,1\},\vee,\wedge,0,1)} .$$

3.2 Maximum calculus

The maximal coordinate of a standard TN is the contraction wrt the semiring

$$(\mathbb{R}, \max, \cdot, 0, 1) .$$

Analogously the minimal coordinate by

$$(\mathbb{R}, \min, \cdot, 0, 1) .$$

These schemes only work, when taking the minimal coordiante keeping all variables open. When searching for the maximal/minimal coordiante of a contraction with some variables closed, we need to first coarse grain the tensor network.

These contractions are used in encoding-decoding schemes Mézard (2009); MacKay (2003).

References

David J. C. MacKay. *Information Theory, Inference and Learning Algorithms*. Cambridge University Press, Cambridge, illustrated edition edition, September 2003. ISBN 978-0-521-64298-9.

Marc Mézard. *Information, Physics, and Computation*. ACADEMIC, Oxford ; New York, March 2009. ISBN 978-0-19-857083-7.